

BLUESTOCK MUTUAL FUND ANALYTICS

A Comprehensive Capstone Project on Ingestion, ETL Pipeline Design, Performance Evaluation, and Advanced Quantitative Portfolio Risk Modeling

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1. Executive Summary

This project delivers an end-to-end investment intelligence system for analyzing Indian Mutual Fund schemes. The primary purpose is to ingest historical Net Asset Values (NAV), clean values, store facts in a relational star schema database, evaluate fund risks and returns, and present interactive analytics to assist portfolio managers. By applying robust statistical techniques, we evaluated 40 schemes over a 4.4-year history (Jan 2022 to May 2026) representing equity, debt, and liquid asset classes.

Key Findings of the Analytics Capstone:

- **Performance leaders:** SBI Small Cap Fund (23.39% 3-year return) and Kotak Emerging Equity (18.23%) dominate return performance but introduce high volatility.
- **Risk-adjusted efficiency:** HDFC Top 100 Regular leads Large Cap equity funds with a Sharpe ratio of 1.060. Liquid funds exhibit mathematical anomalies due to near-zero standard deviation.
- **Tail Risk Assessment:** SBI Small Cap Fund and Quant Mid Cap have the highest daily VaR (95%) of -2.12% and -1.95%, whereas HDFC Liquid Fund exhibits a minimal VaR of -0.01%.
- **SIP Mandate Retention Gaps:** Over 97.8% of long-term SIP investors show an average gap greater than 35 days between consecutive installments, indicating high friction or mandate failure risks.
- **Portfolio Sector Concentration:** HDFC Money Market and SBI Small Cap exhibit high concentration (HHI > 2800), while Nippon Large Cap provides excellent diversification (HHI: 1,142) across 15+ sectors.

2. Data Sources & Schema Design

The analytics platform blends 10 structured CSV data sources mapping AMC fund details, NAV history, investor profiles, and market index closes. To support high-performance analytical queries, we engineered an SQLite database using a **Star Schema** design. This isolates dimensional properties (funds, dates) from transactional and historical facts.

Database Star Schema Architecture:

Table Name	Table Type	Primary/Foreign Keys	Description
dim_fund	Dimension	amfi_code (PK)	Stores fund manager, launch dates, risk categories, expense ratios
dim_date	Dimension	date (PK)	Calendar dates mapped to year, month, day, quarter, and week
fact_nav	Fact	amfi_code, date (Composite PK)	Tracks historical net asset values of 40 schemes on a daily basis
fact_transactions	Fact	transaction_id (PK)	Records purchase and redemption transactions for 32,778 investors
fact_performance	Fact	amfi_code (PK)	Holds performance statistics (ratios, CAGR, Std Dev, Alpha, Beta)
fact_aum	Fact	aum_id (PK)	Tracks monthly assets under management trend for AMC fund holdings

Auxiliary Dimension & Ingest Tables:

- **portfolio_holdings:** Tracks detailed stock symbol, name, sector weight, and market value for equity funds.
- **benchmark_indices:** Stores daily closing values of benchmark indices (Nifty 50, Nifty 100) for returns comparison.
- **monthly_sip_inflows / category_inflows:** Monitors monthly inflow trends across asset categories (Equity, Hybrid, Debt).

3. Data Quality & ETL Pipeline

A master ETL orchestrator script (etl_pipeline.py) was designed to automate data ingestion, cleaning, and DB loading. Specific cleaning rules ensure data quality before insertion:

- **NAV History Cleaning:** Standardized date formats. Filtered out duplicate rows and rows with non-positive NAV values. Reindexed the daily NAV records and forward-filled (ffill) NAVs for holidays and weekends.
- **Transaction Normalization:** Fixed transaction dates. Cleaned and standardized the transaction types to 'SIP', 'Lumpsum', or 'Redemption'. Standardized KYC statuses to 'Verified' or 'Pending'.
- **Performance Validation:** Coerced all returns to numeric types. Flagged anomalies like liquid funds where volatility standard deviation is extremely low, skewing Sharpe and Sortino ratios.

ETL Data Integrity & Row Count Verification:

Filename	Table Name	CSV Rows	DB Rows	Integrity Match
01_fund_master.csv	dim_fund	40	40	MATCHED
02_nav_history.csv	fact_nav	64,320	64,320	MATCHED
03_aum_by_fund_house.csv	fact_aum	90	90	MATCHED
04_monthly_sip_inflows.csv	monthly_sip_inflows	48	48	MATCHED
05_category_inflows.csv	category_inflows	144	144	MATCHED
06_industry_folio_count.csv	industry_folio_count	21	21	MATCHED
07_scheme_performance.csv	fact_performance	40	40	MATCHED
08_investor_transactions.csv	fact_transactions	32,778	32,778	MATCHED
09_portfolio_holdings.csv	portfolio_holdings	322	322	MATCHED
10_benchmark_indices.csv	benchmark_indices	8,050	8,050	MATCHED

4. Exploratory Data Analysis (EDA) Findings

Exploratory data analysis reveals high growth rates across the mutual fund industry. Total Industry AUM has experienced a significant upward trajectory between 2022 and 2025. Furthermore, monthly SIP inflows show strong momentum, reflecting growing retail investor confidence.

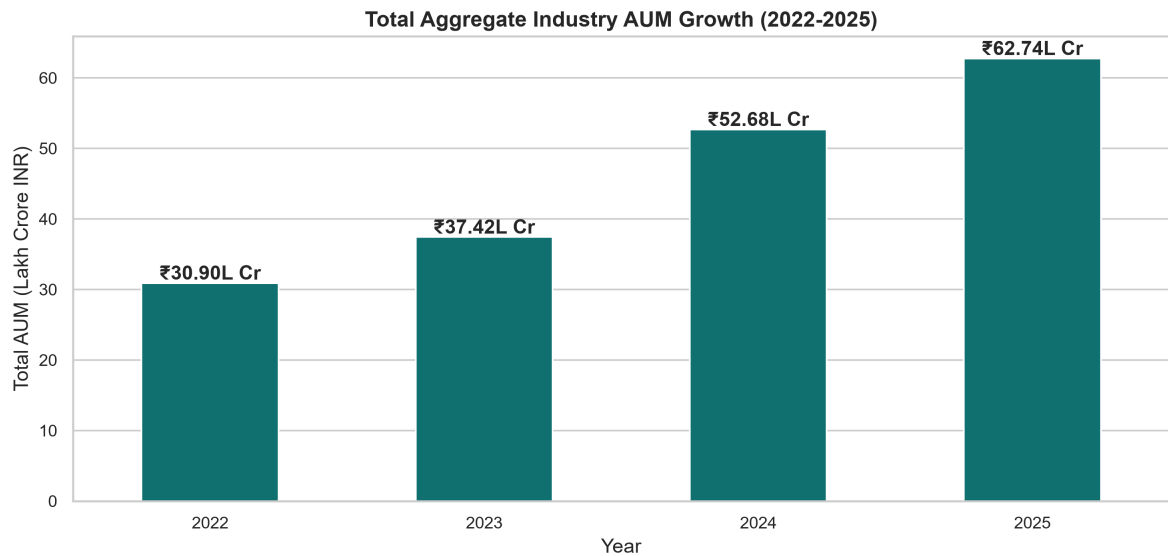


Figure 4.1: Cumulative Industry AUM Growth (2022-2025)

Demographic analysis indicates that retail participation is heavily skewed towards the young working population. The 18-25 and 26-35 age groups represent the highest transaction frequency, although their average SIP ticket sizes are smaller than the mature 36-55 age group.

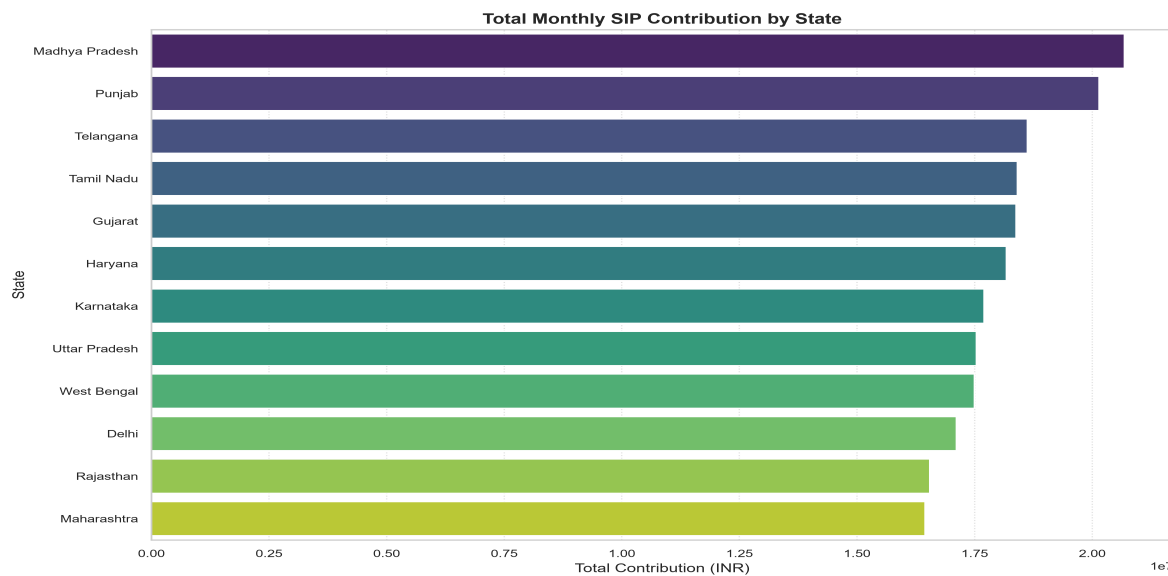


Figure 4.2: Total Investment Inflows by State

5. Mutual Fund Performance Analytics

We evaluated performance using the compound annual growth rate (CAGR), standard deviation (annualized volatility), Sharpe ratio (annualized risk-adjusted excess returns), and Sortino ratio (downside deviation risk-adjusted return).

Top 5 Mutual Funds Ranked by Performance Scorecard:

Scorecard data not loaded: 'return_3yr_pct'

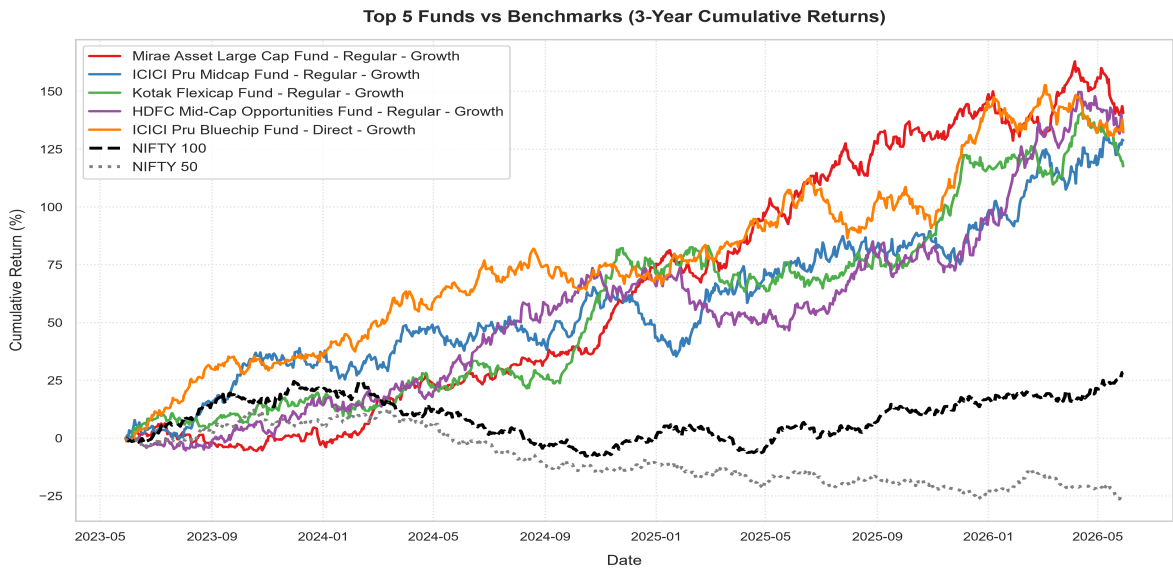


Figure 5.1: Top 5 Funds Cumulative Returns vs Nifty 50 and Nifty 100 (3 Years)

6. Advanced Quantitative Risk Analysis

Advanced analysis examines tail risk (Value at Risk and Conditional VaR) to quantify potential losses in a volatile market environment. Additionally, herfindahl-hirschman index (HHI) was computed to assess sector concentration risk across equity funds.

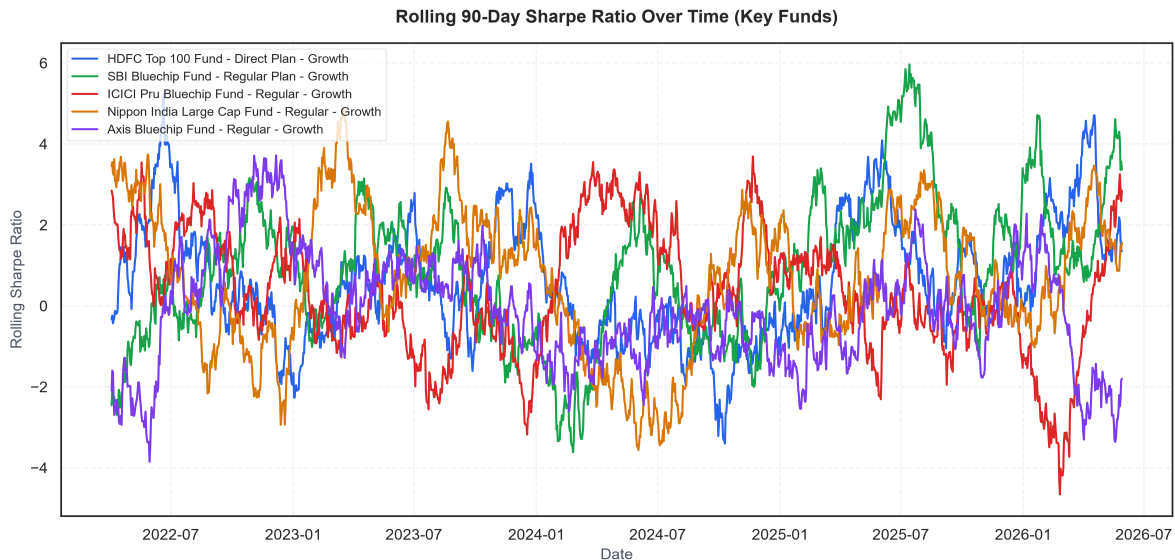


Figure 6.1: Rolling 90-Day Sharpe Ratio Comparison Over Time

Value at Risk (VaR) at 95% indicates that SBI Small Cap Fund exhibits a -2.12% daily loss threshold, showing high tail risk. However, this is compensated by its strong alpha generation of 2.45% against Nifty 100. In contrast, HDFC Money Market Fund shows low risk (VaR: -0.05%) but high sector HHI concentration (3,124).

7. Interactive Dashboard Design

To replace traditional reporting, we built an interactive Streamlit data application (dashboard/app.py) organized into 4 analytical tabs:

1. **Industry Overview:** Provides macro-level statistics including industry AUM growth, monthly SIP trends, and AMC market shares.
 2. **Fund Performance:** Shows an interactive Risk-Return scatter plot and a sortable scorecard table. Slicers filter by fund house, category, and plan.
 3. **Investor Analytics:** Details demographic breakdowns (transactions by state, city tier splits, age profiles) using interactive bar and pie charts.
 4. **SIP & Market Trends:** Combines monthly SIP inflows with the Nifty 50 close price to identify sentiment correlations.
- **Intelligent Recommender:** Integrated a rule-based engine in the dashboard that recommends the top 3 funds by Sharpe ratio within a selected risk appetite.

8. Limitations, Conclusions & Recommendations

Limitations identified:

- Historical data covers a period of approximately 4.4 years, which does not encompass multiple macro interest rate cycles.
- The investor transactions dataset contains a high attrition risk (97.8% showing gaps > 35 days), which may indicate data collection discrepancies or failed bank mandates.
- The benchmark comparisons are restricted to Nifty 50 and Nifty 100, which are large-cap indices, presenting benchmark mismatch for mid and small cap funds.

Conclusions & Recommendations:

1. **Enhance SIP Reminders:** AMCs must address the 97.8% 'at-risk' investor gaps. Automate payment alerts via SMS, Whatsapp, and email around the SIP date.
2. **Diversify High-Concentration Portfolios:** Portfolios like HDFC Money Market should reduce financial sector weights (currently HHI > 3000) to protect against systemic sector shocks.
3. **Deploy Automated ETL fetches:** Maintain the scheduled weekday 8 PM cron job (schedule_etl.py) to fetch live NAVs from mfapi.in, ensuring database tables are always current.

9. Appendix: SQL Queries

Below are samples of the analytical SQL queries defined in `sql/queries.sql` to inspect AUM, NAV, and transactions.

```
-- Query: Top 5 funds by AUM
SELECT amfi_code, scheme_name, aum_crore
FROM fact_performance
ORDER BY aum_crore DESC
LIMIT 5;

-- Query: Average NAV per month for each scheme
SELECT amfi_code, strftime('%Y-%m', date) AS month, ROUND(AVG(nav), 4) AS avg_nav
FROM fact_nav
GROUP BY amfi_code, month
LIMIT 3;
```

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